UNDERSTANDING CUSTOMER SPENDING TRENDS FOR SMARTER BUSINESS DECISIONS

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CLEANING

```
# Import the dplyr library
library(dplyr)
```

```
# Preparing packages - dplyr
```

```
# The following objects are masked from 'package:stats':
#
#     filter, lag
#
# The following objects are masked from 'package:base':
#
#     intersect, setdiff, setequal, union
```

```
# Load the ggplot2 library
library(ggplot2)
```

```
# Warning: package 'ggplot2' was built under R version 4.0.3
```

```
# Read the CSV file into a data frame
data <- read.csv("/kaggle/input/Shopping Trends.csv")

# Display the first 10 rows of the data
head(data, 10)
```

```
  customer_id Age Gender Item.Purchased Category Purchase.Amount..USD.
1           1  55   Male       Blouse  Clothing                   53
2           2  19 Male       Sweater  Clothing                   64
3           3  50 Male      Jeans  Clothing                   73
4           4  21 Male    Sandals Footwear                   90
5           5  45 Male     Blouse  Clothing                   49
6           6  46 Male     Sweater  Clothing                   20
7           7  63 Male      Jeans  Clothing                   85
8           8  27 Male   Kasa Hat Accessories                   34
9           9  26 Male     Jacket  Clothing                   73
10         10  57 Male     Belt Accessories                   90
   Location Size Color Season Review.Rating Subscribed.Status
1  Kentucky    L  Gray Winter             3            Yes
2    Maine    L Black Winter             3             No
3 Massachusetts    S Teal Winter             2            Yes
4 Rhode Island    M Green Spring             3             No
5 Pennsylvania    L Gray Spring             3             No
6  Arkansas    S Green Summer             4             No
7   Hawaii    M Black Summer             4             No
8 Maryland    S Green Summer             3             No
9 Maryland    S Green Summer             3             No
10 Massachusetts    M Teal Summer             3             No
   Shipping.Type Discount.Applied Promo.Code.Used Previous.Purchases
1       Express             Yes             Yes                 14
2       Express             Yes             Yes                  2
3       Express             Yes             Yes                 23
4       Express             Yes             Yes                 49
5       Express             Yes             Yes                 31
6       Express             Yes             Yes                  7
7       Express             Yes             Yes                 32
8       Express             Yes             Yes                 44
9       Express             Yes             Yes                 44
10      Express             Yes             Yes                 41
   Payment.Method Frequency.of.Purchases
1         Venmo       Fortnightly
2          Cash            Weekly
3        Credit       Weekly
4          PayPal       Weekly
5        Credit       Weekly
6        Credit       Weekly
7        Credit       Weekly
8        Credit       Weekly
9        Credit       Weekly
10         Venmo       Weekly
```

```
# Check for missing values in the data
sum(is.na(data))
```

```
[1] 0
```

```
# Check for duplicate rows
sum(duplicated(data))
```

```
[1] 0
```

```
# Convert categorical variables to factors
data$Gender <- as.factor(data$Gender)
data$Category <- as.factor(data$Category)
data$Season <- as.factor(data$Season)
data$Subscribed.Status <- as.factor(data$Subscribed.Status)
data$Discount.Applied <- as.factor(data$Discount.Applied)
data$Shipping.Type <- as.factor(data$Shipping.Type)

# Summary statistics for numerical variables
summary(data$Age)
summary(data$Purchase.Amount..USD.)
summary(data$Review.Rating)

# Frequency tables for categorical variables
table(data$Gender)
table(data$Category)
table(data$Season)
table(data$Subscribed.Status)
table(data$Discount.Applied)
table(data$Shipping.Type)

# Correlation between numerical variables
cor(data$Age, data$Purchase.Amount..USD.)
cor(data$Age, data$Review.Rating)
cor(data$Purchase.Amount..USD., data$Review.Rating)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   18.0    34.0    44.0    44.1    54.0    70.0
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     20      39      60      60      81     100
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    2.5     3.2     3.7     3.7     4.2     5.0

  Female   Male
    1248   2652

Accessories    Clothing    Footwear   Outerwear
        599        1737         324         155

     Fall    Spring    Summer    Winter
     1035       999       957       952

     No    Yes
   2847   1053

     No    Yes
   2223   1677

Express   Free Next Day   Second Day   Standard   Store Pickup
    646        655        675       1842        654

[1] 0.0372312

[1] -0.0312683

[1] -0.0342141
```

EXPLORATORY DATA ANALYSIS

```
# Load the ggplot2 library
library(ggplot2)
```

```
# Warning: package 'ggplot2' was built under R version 4.0.3
```

```
# Create a boxplot of purchase amount by gender
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Average Purchase Amount by Gender")
  + xlab("Gender")
  + ylab("Purchase Amount (USD)")
  + scale_fill_manual(values=c("#ff0000", "#00ff00", "#0000ff"))
  + theme_minimal()
```

```
# Create a boxplot of purchase amount by subscription status
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Average Past Purchases by Subscription Status")
  + xlab("Subscribed")
  + ylab("Avg Previous Purchases")
  + scale_fill_manual(values=c("#ff0000", "#00ff00"))
  + theme_minimal()
```

```
# Create a boxplot of purchase amount by discount applied
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Impact of Discount Application on Purchase Amount")
  + xlab("Discount Applied")
  + ylab("Purchase Amount (USD)")
  + scale_fill_manual(values=c("#ff0000", "#00ff00"))
  + theme_minimal()
```

```
# Create a boxplot of shipping type by purchase amount
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Shipping Type Popularity")
  + xlab("Shipping Type")
  + ylab("Purchase Amount (USD)")
  + coord_flip()
  + theme_minimal()
```

```
# Create a boxplot of purchase amount by frequency of purchases
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Purchase Frequency by Age Group")
  + xlab("Age Group")
  + ylab("Purchase Amount (USD)")
  + scale_fill_manual(values=c("#ff0000", "#00ff00", "#0000ff", "#ffff00", "#ff00ff", "#00ffff", "#ff7f00"))
  + theme_minimal()
```

```
# Create a heatmap of correlations between numerical variables
library(ggplot2)
library(reshape2)
cor_matrix <- cor(data[c("Age", "Previous.Purchases", "Review.Rating", "Purchase.Amount..USD.")])
melted_cor_matrix <- melt(cor_matrix)
ggplot(melted_cor_matrix, aligned=TRUE)
  + geom_tile(aligned=TRUE)
  + ggtitle("Correlation Heatmap")
  + xlab("")
  + ylab("")
  + scale_fill_gradient(low="#ffffff", high="#ff0000")
  + theme_minimal()
```

```
# Warning: package 'reshape2' was built under R version 4.0.3
```

```
# Create a boxplot of sales by season and product category
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Seasonal Sales by Product Category")
  + xlab("Season")
  + ylab("Purchase Amount (USD)")
  + scale_fill_manual(values=c("#ff0000", "#00ff00", "#0000ff", "#ffff00"))
  + theme_minimal()
```



```
# Create a boxplot of average purchase amount by product category
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Average Purchase by Product Category")
  + xlab("Avg_Purchase")
  + ylab("Product Category")
  + scale_fill_manual(values=c("#ff0000", "#00ff00", "#0000ff", "#ffff00"))
  + theme_minimal()
```

```
# Create a boxplot of average purchase by season
ggplot(data, aligned=TRUE)
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Average Purchase by Season")
  + xlab("Season")
  + ylab("Avg_Purchase")
  + scale_fill_manual(values=c("#ff0000", "#00ff00", "#0000ff", "#ffff00"))
  + theme_minimal()
```


1. Average Purchase Amount by Gender Observation: The average purchase amount by female customers is slightly higher than that of male customers. Interpretation: While both genders spend almost equally, small behavioral differences might be leveraged in personalized marketing.
2. Average Previous Purchases by Subscription Status Observation: Customers with an active subscription ("Yes") have a higher average of previous purchases than unsubscribed ones. Interpretation: Subscription is a loyal trend to customer loyalty or repeat purchase.
3. Discount Applied vs Purchase Amount Observation: Customers who received a discount spend more on average than those who did not. Interpretation: Discounts effectively boost purchase amounts, suggesting they are a strong tool for increasing sales and encouraging bulk purchases.
4. Shipping Type Popularity Observation: Standard shipping is the most popular, with Free Shipping being slightly more preferred. Interpretation: Providing free shipping or offering it as a premium option could be a strategic consideration for value-driven options.
5. Purchase Frequency by Age Group Observation: Older age groups (46-70) tend to shop more frequently across all frequencies. Younger customers (18-35) show more variability in their purchasing frequency, with some exhibiting more frequent shopping habits for disposable income. Younger demographics might require more engagement or incentives.
6. Correlation Heatmap Observation: We made correlations between Purchase, Amount and all other numeric variables. Interpretation: No strong correlations between numerical variables.
7. Seasonal Sales by Product Category Observation: Clothing dominates all seasons. Accessories perform second-best and peak in Summer. Footwear and Outerwear show lower sales, with Outerwear slightly higher in Winter. Interpretation: Clothing is the most popular product category. For example, stock more clothing during the year, with accessories in Fall/Summer.

QUESTION 1: What factors influence the amount a customer spends in a single transaction?

```
# Regression model to predict purchase amount
# Using Age, Gender, Previous Purchases, Review Rating, and Subscribed Status as predictors
model_formula <- Purchase.Amount..USD. ~ Age + Gender + Previous.Purchases + Review.Rating + Subscribed.Status
model <- lm(model_formula, data=data)
summary(model)

# Interpretation:
# Coefficients: Shows the impact of each predictor on purchase amount.
# Significance: Check p-values to see if the predictors are statistically significant.
# Model fit: Check R-squared and adjusted R-squared to see how well the model explains the data.

# Create a scatter plot of Age vs Purchase Amount, colored by Gender
ggplot(data, aes(x=Age, y=Purchase.Amount..USD., color=Gender))
  + geom_point()
  + ggtitle("Age vs Purchase Amount by Gender")
  + xlab("Age")
  + ylab("Purchase Amount (USD)")
  + scale_color_manual(values=c("#ff0000", "#00ff00", "#0000ff"))
  + theme_minimal()

# Create a boxplot of Previous Purchases by Subscribed Status
ggplot(data, aes(x=Subscribed.Status, y=Previous.Purchases))
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Previous Purchases by Subscribed Status")
  + xlab("Subscribed Status")
  + ylab("Previous Purchases")
  + scale_fill_manual(values=c("#ff0000", "#00ff00"))
  + theme_minimal()

# Create a boxplot of Review Rating by Subscribed Status
ggplot(data, aes(x=Subscribed.Status, y=Review.Rating))
  + geom_boxplot(aligned=TRUE)
  + ggtitle("Review Rating by Subscribed Status")
  + xlab("Subscribed Status")
  + ylab("Review Rating")
  + scale_fill_manual(values=c("#ff0000", "#00ff00"))
  + theme_minimal()
```

```
Call:
lm(formula = model_formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-  -  -  -
```

```
{"id": 1270, "name": "Ellie Jones", "email": "ellie.jones@yahoo.com", "phone": "(555)-234-5678", "address": "789 Oak Ave, Springfield, IL 62704", "publication_date": "2020-08-15", "publication_time": "14:30:00", "publication_datetime": "2020-08-15T14:30:00", "publication_timestamp": 1597499400, "publication_year": 2020, "publication_month": 8, "publication_day": 15, "publication_hour": 14, "publication_minute": 30, "publication_second": 0, "publication_weekday": 6, "publication_weekday_name": "Saturday", "publication_week": 33, "publication_quarter": 3, "publication_decade": 2020, "publication_century": 21, "publication_millennium": 3, "publication_iso_week": "2020-W33", "publication_iso_weekday": 6, "publication_iso_year": 2020, "publication_iso_month": 8, "publication_iso_day": 15, "publication_iso_hour": 14, "publication_iso_minute": 30, "publication_iso_second": 0, "publication_iso_timestamp": 1597499400, "publication_iso_datetime": "2020-08-15T14:30:00", "publication_iso_date": 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```

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# ---- Ran 1 test in 0.000s
# Ran 1 test in 0.000s
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Interpretations for Question 1: To explore the factors that influence how much a customer spends in a single transaction, we began by fitting a linear regression model using demographic and behavioral features such as Age, Gender, Review Rating, Subscription Status, Discount Applied, Previous Purchases, and Shipping Type. The model achieved an R² of 0.0, indicating that none of these variables explained any significant spending variance. Using stepwise AIC selection, Other variables such as Review Rating, Payment Method, Gender, Credit Card, and Item Purchased were retained in the final model, but their combined effect was still small (R² = 0.0001), suggesting that these relationships are statistically significant but practically negligible.

Interpretations for Question 2: To explore the factors that influence how much a customer spends in a single transaction, we began by fitting a linear regression model. Demographic and behavioral features such as Age, Gender, Review Rating, Subscription Status, Discount Applied, Previous Purchases, and Shipping Type were used to predict Purchase Amount (USD). The model explained a very small portion of the variance (R² = 0.0), indicating that none of these variables had a strong and consistent linear relationship with spending. However, several variables were statistically significant, including Review Rating, Payment Method, Gender, Credit Card, and Item Purchased. Although the effect sizes were small, these factors still contribute to spending behavior in ways that may be worth exploring in more detailed analysis. Interactions between variables and may be better captured using tree-based models.

QUESTION 1: Can we identify patterns in repeat purchases based on customer demographics and shopping preferences?

```
# Load customer transaction data into a dataframe.
dataset$Repeat_Customer <- ifelse(dataset$Previous_Purchases > 1, 1, 0)
table(dataset$Repeat_Customer)
prop.table(table(dataset$Repeat_Customer))

table
  0   1
1948 2552

0.4416667 0.5783333
```

```
# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a bar chart.
ggplot(data = dataset, aes(x = factor(Repeat_Customer)))
  + geom_bar(fill = c("#ff6666", "#66cc99"))
  + labs(title = "Repeat Customer Distribution",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Subscribed_Customer)))
  + labs(title = "Repeat Customer Distribution by Subscribed Customer",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Subscribed Customer")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Discount_Applied)))
  + labs(title = "Repeat Customer Distribution by Discount Applied",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Discount Applied")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Review_Rating)))
  + labs(title = "Repeat Customer Distribution by Review Rating",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Review Rating")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Payment_Method)))
  + labs(title = "Repeat Customer Distribution by Payment Method",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Payment Method")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Item_Purchased)))
  + labs(title = "Repeat Customer Distribution by Item Purchased",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Item Purchased")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Shipping_Type)))
  + labs(title = "Repeat Customer Distribution by Shipping Type",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Shipping Type")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Credit_Card)))
  + labs(title = "Repeat Customer Distribution by Credit Card",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Credit Card")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Gender)))
  + labs(title = "Repeat Customer Distribution by Gender",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Gender")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Age)))
  + labs(title = "Repeat Customer Distribution by Age",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Age")
+ theme_minimal()
```

QUESTION 2: Can we identify patterns in repeat purchases based on customer demographics and shopping preferences?

```
# Load customer transaction data into a dataframe.
dataset$Repeat_Customer <- ifelse(dataset$Previous_Purchases > 1, 1, 0)
table(dataset$Repeat_Customer)
prop.table(table(dataset$Repeat_Customer))

table
  0   1
1948 2552

0.4416667 0.5783333
```

```
# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a bar chart.
ggplot(data = dataset, aes(x = factor(Repeat_Customer)))
  + geom_bar(fill = c("#ff6666", "#66cc99"))
  + labs(title = "Repeat Customer Distribution",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Subscribed_Customer)))
  + labs(title = "Repeat Customer Distribution by Subscribed Customer",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Subscribed Customer")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Discount_Applied)))
  + labs(title = "Repeat Customer Distribution by Discount Applied",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Discount Applied")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Review_Rating)))
  + labs(title = "Repeat Customer Distribution by Review Rating",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Review Rating")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Payment_Method)))
  + labs(title = "Repeat Customer Distribution by Payment Method",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Payment Method")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Item_Purchased)))
  + labs(title = "Repeat Customer Distribution by Item Purchased",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Item Purchased")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Shipping_Type)))
  + labs(title = "Repeat Customer Distribution by Shipping Type",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Shipping Type")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Credit_Card)))
  + labs(title = "Repeat Customer Distribution by Credit Card",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Credit Card")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Gender)))
  + labs(title = "Repeat Customer Distribution by Gender",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Gender")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Age)))
  + labs(title = "Repeat Customer Distribution by Age",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Age")
+ theme_minimal()
```

QUESTION 3: Can we predict if a customer will subscribe based on their purchase behavior?

```
# Load customer transaction data into a dataframe.
dataset$Repeat_Customer <- ifelse(dataset$Previous_Purchases > 1, 1, 0)
table(dataset$Repeat_Customer)
prop.table(table(dataset$Repeat_Customer))

table
  0   1
1948 2552

0.4416667 0.5783333
```

```
# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a bar chart.
ggplot(data = dataset, aes(x = factor(Repeat_Customer)))
  + geom_bar(fill = c("#ff6666", "#66cc99"))
  + labs(title = "Repeat Customer Distribution",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Subscribed_Customer)))
  + labs(title = "Repeat Customer Distribution by Subscribed Customer",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Subscribed Customer")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Discount_Applied)))
  + labs(title = "Repeat Customer Distribution by Discount Applied",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Discount Applied")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Review_Rating)))
  + labs(title = "Repeat Customer Distribution by Review Rating",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Review Rating")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Payment_Method)))
  + labs(title = "Repeat Customer Distribution by Payment Method",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Payment Method")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Item_Purchased)))
  + labs(title = "Repeat Customer Distribution by Item Purchased",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Item Purchased")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Shipping_Type)))
  + labs(title = "Repeat Customer Distribution by Shipping Type",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Shipping Type")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Credit_Card)))
  + labs(title = "Repeat Customer Distribution by Credit Card",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Credit Card")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Gender)))
  + labs(title = "Repeat Customer Distribution by Gender",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Gender")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a stacked bar chart.
ggplot(data = dataset, aligned = TRUE)
  + geom_bar(aes(x = factor(Repeat_Customer), fill = factor(Age)))
  + labs(title = "Repeat Customer Distribution by Age",
         x = "Repeat Customer (0 = No, 1 = Yes)",
         y = "Count",
         fill = "Age")
+ theme_minimal()
```

QUESTION 4: Can we cluster customers based on shopping behavior to identify personas?

```
# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Review_Rating, y = Previous_Purchases))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Review Rating vs. Previous Purchases",
         x = "Review Rating",
         y = "Previous Purchases")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Age, y = Previous_Purchases))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Age vs. Previous Purchases",
         x = "Age",
         y = "Previous Purchases")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Purchase_Amount_(USD), y = Previous_Purchases))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Purchase Amount (USD) vs. Previous Purchases",
         x = "Purchase Amount (USD)",
         y = "Previous Purchases")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Review_Rating, y = Purchase_Amount_(USD)))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Review Rating vs. Purchase Amount (USD)",
         x = "Review Rating",
         y = "Purchase Amount (USD)")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Age, y = Purchase_Amount_(USD)))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Age vs. Purchase Amount (USD)",
         x = "Age",
         y = "Purchase Amount (USD)")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Previous_Purchases, y = Purchase_Amount_(USD)))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Previous Purchases vs. Purchase Amount (USD)",
         x = "Previous Purchases",
         y = "Purchase Amount (USD)")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Age, y = Review_Rating))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Age vs. Review Rating",
         x = "Age",
         y = "Review Rating")
+ theme_minimal()
```

QUESTION 5: Can we predict customer lifetime value?

```
# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Subscription_Status, y = Purchase_Amount_(USD)))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Subscription Status vs. Purchase Amount (USD)",
         x = "Subscription Status",
         y = "Purchase Amount (USD)")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Subscription_Status, y = Previous_Purchases))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Subscription Status vs. Previous Purchases",
         x = "Subscription Status",
         y = "Previous Purchases")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Subscription_Status, y = Review_Rating))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Subscription Status vs. Review Rating",
         x = "Subscription Status",
         y = "Review Rating")
+ theme_minimal()

# Install and load the ggplot2 package.
install.packages("ggplot2")
library(ggplot2)

# Create a scatterplot.
ggplot(data = dataset, aes(x = Subscription_Status, y = Age))
  + geom_point(color = "#1f78b4")
  + labs(title = "Scatterplot: Subscription Status vs. Age",
         x = "Subscription Status",
         y = "Age")
+ theme_minimal()
```

Interpretations for Question 3: To uncover meaningful customer segments, we applied K-Means clustering on five key behavioral features: Age, Review Rating, Purchase Amount (USD), and Previous Purchases. The algorithm grouped customers into three distinct clusters, each representing a unique persona based on purchasing behavior and engagement. To visualize and validate these clusters, we used Principal Component Analysis (PCA) to reduce the data to two dimensions, allowing us to plot the customer groups in a scatterplot. The first two principal components, accounting for the most variance, corresponded to the clusters. This approach ensured that the segmentation was both statistically robust and visually interpretable, providing a clear foundation for targeted marketing strategies and personalized customer campaigns.

Based on this analysis, we identified three customer personas: 1. Value Shoppers – Lower spend and fewer purchases 2. Loyal Mid-Spenders – Moderate spend and moderate loyalty 3. High-Value Loyalists – Frequent purchasers with high review ratings and strong purchase volume

These insights provide a strong foundation for segment-specific strategies, personalized offers, and customer retention campaigns tailored to each group's behavior and value.